

# **Extract, then decide: a two-stage pipeline to identify seizure frequency in epilepsy clinic letters**

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Conor Brown  
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## Declaration of Academic Integrity

I certify that the submission is my own work, all sources are correctly attributed, and the contribution of any AI technologies is fully acknowledged.

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A handwritten signature in black ink. The word "Brown" is written in a bold, slightly cursive font. It is enclosed within a large, sweeping, horizontal loop that starts above the first letter, goes under the entire word, and ends above the last letter.

Student's signature:

Date of submission: 4th September 2026

# Extract, then decide: a two-stage pipeline to identify seizure frequency in epilepsy clinic letters

**Abstract**—Large language models can extract seizure frequency patterns from epilepsy letters. However, fine-tuning a model for this task requires substantial data and computation. This paper proposes a combination of prompt engineering and rule-based systems as an alternative to fine-tuning. It uses a two-stage pipeline to extract a rich structured evidence record, then decide the final label using a fixed policy. The model extracts all seizure frequency patterns in the text, quotes exact evidence for each and makes an initial proposal for the patient’s current status. A subsequent model call or rules-based system reviews that evidence against a defined policy and provides the final answer. The pipeline was developed using a dataset of 1,200 synthetic letters. Performance was evaluated on a clinician double-checked set of 300 real clinic letters and compared against a fine-tuned model previously evaluated on the same test set. The two-stage pipeline delivered comparable performance (micro-F1: 0.77 and 0.80) to the fine-tuned model (0.79). Models can reliably extract multiple seizure frequency patterns with exact evidence. Rules can reliably use that evidence to determine the correct answer in single-label extraction tasks. This method produces a richer evidence chain with comparable accuracy.

**Index Terms**—information extraction, hybrid pipeline, large language models, seizure frequency, epilepsy

## I. INTRODUCTION

Epilepsy affects c. 1% of the UK population [1] and is characterised by recurrent, unprovoked seizures [2]. Seizure frequency is a key indicator of the patient’s disease burden and is commonly used in clinical care to determine the appropriate treatment and as a primary outcome in clinical trials. Seizure frequency is routinely recorded in clinic letters, unstructured narrative reports of a patient’s recent visit to the epilepsy clinic, mixed in with descriptions of past diagnoses, current symptoms and future treatment. This information can be used to conduct retrospective studies or to identify patient cohorts for clinical trials, but extracting it into that structured form requires experts to conduct costly, error-prone manual chart review.

Natural language processing (NLP) offers an automated alternative, but clinical narrative presents distinct challenges. Traditional rule-based systems provide deterministic, auditable logic valued by clinicians, but they struggle with the lexical variety and syntactic complexity of clinical frequency descriptions, which range from explicit rates (*2 to 3 seizures a week*) to relative interval counts (*4 since her last visit*) and qualitative impressions (*frequent bouts*) [3]. Conversely, large language models (LLMs) demonstrate remarkable semantic flexibility in zero- and few-shot clinical information extraction [4], [5]. However, unconstrained LLMs introduce substantial clinical risks: they operate as black boxes, remain prone to

hallucinations or ungrounded assertions [6], and existing state-of-the-art results have relied on fine-tuning on large collections of expert-annotated letters [3].

Seizure frequency extraction is framed as a single-label extraction task in prior work [3], but it consists of multiple sub-tasks. A clinic letter can describe several seizure events of different types and on different occasions, so the system must first extract every candidate fact with its evidence, then decide which candidate, or which combination of them, best represents the patient’s current seizure frequency burden. Which to keep is a clinical policy that varies between users. Keeping the two stages apart captures more evidence and makes the final answer explainable, as the end user can see the candidates, the policy and the label it produced.

The primary objective of this paper is to evaluate whether the two-stage extraction pipeline using prompt engineering and rules-based methods can extract seizure frequency patterns from epilepsy letters with comparable accuracy to fine-tuned models [3]. The secondary objective is to evaluate whether the decide stage improves on the provisional answer from the initial extract stage. The decide stage is implemented as both a second LLM call (LLM-only) and a rules-based system (Hybrid) and both methods are evaluated. If both objectives are met, the paper contributes a new method for seizure frequency extraction.

To answer these questions, the pipeline was developed on 1,200 synthetic epilepsy clinic letters produced by Gan *et al.* [3], each with a single expert-annotated label for the patient’s current seizure frequency pattern. Results are compared with the benchmark Gan *et al.* report for their fine-tuned LLM using the same set of 300 clinic letters from real patients. The primary evaluation metric is the F1 score.

The rest of the paper is organised as follows. First, the literature review summarises previous methods used to extract seizure frequency patterns from epilepsy letters, compares conceptual framings on the different sub-tasks of clinical IE and establishes how this method is different. Next, the methods section describes the dataset used for evaluation, the overall architecture of the pipeline and the specific implementation details of the LLM and rules. Next, the results section presents the results of the experiments, comparing the method against past benchmarks and examining its strengths and weaknesses. Next, the discussion section presents implications of the results along with the limitations of the work and ethical considerations. Finally, the conclusion section summarises the key findings and highlights opportunities for future research.

## II. LITERATURE REVIEW

The best approach to extract seizure frequency information from epilepsy clinic letters is an active area of research, with experiments using rules-based methods [7], [8], transformer-based methods [3], [6], [9], [10] and a combination of both [11], [12] all published this decade. Rules-based methods struggled with the lexical variety of seizure frequency patterns. A key challenge with both rules-based methods and BERT-based methods was their failure to generalise to letters from a different hospital [8], [12]. LLMs were the first method to demonstrate strong generalisation to a new dataset [3], but this was achieved by fine-tuning a model on over 1,000 synthetic letters. This paper explores whether similar performance can be achieved with a two-stage extraction pipeline.

Clinical IE has been defined as the automatic extraction and encoding of clinical information from text [13]: recognising a mention and normalising it into a standard form [14]. Modern LLMs do both in one step when asked for a normalised label such as *4 per month* from a raw statement such as *about once a week*. A further step exists in many clinical extraction pipelines but goes unmentioned: deciding between candidates. In a clinic letter with several seizure frequency statements, producing a single label for the patient’s current status requires the system both to extract the relevant statements and to decide which one best answers the question. Prior rules-based systems made this a distinct stage [15]; LLM pipelines instead make the decision inside the extraction prompt. This paper explores whether separating *extract* from *decide* in a two-stage pipeline can deliver comparable performance while preserving more of the evidence for downstream use.

Academic papers tend to focus exclusively on benchmark accuracy while practitioners tend to value transparency, configurability and efficiency at the cost of some accuracy [15], [16]. Model interpretability is also a critical factor in clinical settings [14]. Clinicians are trained to use deductive logic to draw conclusions and they expect systems to expose transparent, interpretable evidence that they can reliably use to make decisions that demand high accountability. Past work utilises evidence spans as a training input [3] for fine-tuning or as data exhaust as part of using a rule-based method [7]. This paper explores whether preserving a transparent evidence chain from exact text to final answer can make the output both more interpretable and more accurate.

## III. METHODS

### A. Data

The dataset is a collection of epilepsy clinic letters which record a UK patient’s consultation with an epilepsy specialist, often a neurologist or specialist nurse [3]. The NHS requires that clinic letters are sent to the patient’s GP after all outpatient visits such as these. They are often the only record of a patient’s diagnoses, treatments and outcomes in neurological disorders, and they are written in narrative form. An example clinic letter is shown in Figure 1.

Gan *et al.* [3] produced this dataset of synthetic letters modelled on real clinic letters from King’s College London

Thank you for asking me to review your patient. I am concerned that her epilepsy has worsened in recent months, with a clear pattern of deterioration. She and her partner report increasing clusters of events correlating with the perimenstrual period, suggesting a catamenial component. In particular, she describes brief focal episodes with preserved awareness that tend to occur as she is settling to sleep, at the point of drifting off, now happening on most nights of the week. These events are accompanied by a rising epigastric sensation, facial pallor, and right-hand automatisms lasting 30–60 seconds, followed by fatigue and headache. In addition, she has had three generalised tonic–clonic seizures in the last six weeks, two of which occurred during the same perimenstrual window.

Fig. 1: An example clinic letter with two distinct seizure frequency patterns highlighted as potential candidates.

TABLE I: Dataset summary

|                   |              | Development | Test |
|-------------------|--------------|-------------|------|
| Number of letters |              | 750         | 450  |
| Seizure frequency | Frequent     | 51%         | 50%  |
|                   | Infrequent   | 17%         | 18%  |
|                   | Unknown      | 17%         | 17%  |
|                   | Seizure free | 15%         | 15%  |

hospital. The real letters were used to create 10 template letters and they were combined with 500 short descriptions of seizure frequency patterns produced by GPT-5 to create the full set of 15,099 letters. A sample of 1,200 letters was used in this paper. Trained experts annotated each letter with a single gold-standard label representing the patient’s current seizure frequency status.

Minimal preprocessing was applied: letters were cleaned only to parse quotes, new lines, tabs, the  $\leq$  sign and nulls, and were ingested in full with no tokenisation, section-splitting or truncation.

The dataset was split into a development set of 750 letters and a test set of 450 letters, stratified by their seizure frequency category (frequent, infrequent, unknown, seizure free) to balance the distribution across both splits. The exact breakdown can be seen in Table I.

Gan *et al.* [3] also used a separately annotated set of 300 real patient clinic letters from King’s College Hospital. This dataset was used to evaluate the pipeline against the previous fine-tuned model.

### B. Architecture

The pipeline is composed of two stages, *extract* and *decide*. The pipeline receives the epilepsy letter in unstructured text as input and the output is a structured evidence record in JSON. The *extract* stage builds the evidence record and the *decide* stage makes decisions based on it. Figure 2 traces the path of a single letter through the pipeline.

The evidence record contains the final answer to the question: *what is the patient’s current seizure frequency burden?*

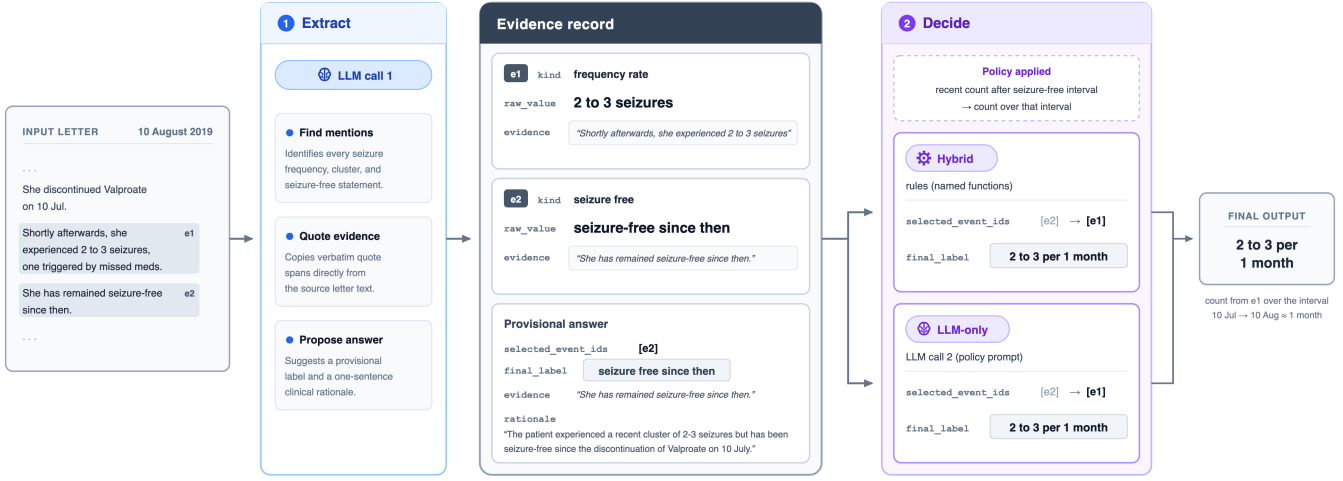


Fig. 2: The pipeline on a development-set letter. The letter narrows to a record and the record to a label: on the development set the extraction call turns a letter of about 400 words into a record of 2.2 events on average (median 2, range 1–6) holding 22 quoted words, and decide returns a label of about four words.

Crucially, it also contains the evidence gathered to determine that answer: multiple seizure frequency statements, exact citations from the text plus normalised labels, and the rationale for choosing that particular seizure frequency pattern.

Extract is an LLM call. It receives the full letter as input along with a detailed set of prompt instructions. The prompt instructions describe the seizure frequency task, the required output schema and examples of how to complete the task. It generates the evidence record illustrated in Table II.

Decide can be implemented as a second LLM call (LLM-only) or a set of rules (Hybrid). Both pipelines share the same extract stage, operate on the same structured evidence, and apply the same set of procedures. Direct comparisons between the two methods measure how effectively they apply the decision policy.

### C. Extract

Extract is the most consequential stage. It ingests the full letter, generates the evidence record and provides the provisional answer. The prompt has four components:

- 1) *Instructions*: find every seizure frequency statement, represent each as an event, assign it one of the six categories in Table II, and propose a provisional answer for the patient’s current seizure frequency.
- 2) *Examples* of how events should be represented.
- 3) *Allowed labels*: a closed list of the canonical label forms used by the gold standard (*N per unit*, *N per N unit*, *seizure free for N unit*, etc.). Each allowed label comes with examples and rules for writing them: digits not words, a bound such as *up to four per day* flattened to *4 per day*, etc.
- 4) An *evidence obligation*: the evidence fields of Table II, with the instruction that each must be an exact quote from the letter.

TABLE II: The evidence record: output of extract, input to decide

| Field                                                          | Content                                                                                           | Example Value                                                           |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <i>One event per seizure frequency statement in the letter</i> |                                                                                                   |                                                                         |
| event_id                                                       | Identifier used by decide                                                                         | e1                                                                      |
| kind                                                           | Frequency rate, cluster frequency, seizure free, last event only, unknown frequency, no reference | frequency rate                                                          |
| raw_value                                                      | The stated rate, duration or last-event statement                                                 | 2 to 3 seizures                                                         |
| evidence                                                       | Exact quote from the letter                                                                       | <i>Shortly afterwards, she experienced 2 to 3 seizures</i>              |
| <i>The provisional answer, one per letter</i>                  |                                                                                                   |                                                                         |
| selected_event_ids                                             | The event or events answered from                                                                 | [e2]                                                                    |
| final_label                                                    | The label                                                                                         | seizure free since then                                                 |
| evidence                                                       | Exact quote supporting the answer                                                                 | <i>She has remained seizure-free since then</i>                         |
| rationale                                                      | One sentence of reasoning                                                                         | <i>... has been seizure-free since the discontinuation of Valproate</i> |

The model therefore does three things in one call: it finds the relevant statements and quotes them, it normalises each into an allowed value, and it chooses among the events. Each of these prompt components is removed in ablations in the results section to evaluate their impact. The full prompt, category definitions and label forms are in the supporting materials.

The type of data captured in the extract stage is shown in Table II. The final\_label is the provisional answer provided by the LLM and it serves as a single prompt baseline

TABLE III: Decision policy: when the final answer differs from the provisional answer

| The record contains                                               | Final answer             |
|-------------------------------------------------------------------|--------------------------|
| An overall count and a breakdown by seizure type                  | The overall count        |
| A seizure-free statement while other seizure-like events continue | The continuing events    |
| A year-to-date total and a typical rate                           | The typical rate         |
| Counts for named months                                           | Total over those months  |
| Seizures on separate dates or months                              | Count over that span     |
| A recent count after a stated seizure-free interval               | Count over that interval |

in this paper. That single label is directly equivalent to the full output by single-stage LLM pipelines in past papers [3]. What distinguishes this pipeline from previous work is that it also records every other seizure frequency pattern in the text.

#### D. Decide

Decide is a smaller and narrower task. Its input is the evidence record; it does not see the full letter and cannot generate new evidence. It keeps the provisional answer unless a rule in the fixed decision policy applies, in which case it either chooses a different event from the record or writes a new label from several of them. Table III lists the policy. In Figure 2 the initial extraction proposed *seizure free* based on the recent seizure free period, but the policy overrides this and chooses the cluster of 2-3 seizures just before it.

In Hybrid, each rule is a named function with regex matching and if-then logic that reads the record. A rule either selects a different already-extracted event or derives a new label from several events with date and numeric logic.

In LLM-only, each rule is a written instruction with one worked example: a small record, the first choice, and the expected answer. The prompt tells the model to keep the first choice unless a row applies, to write a new label only when no single event is the answer, and never to add events or quotes. It returns the selection block of Table II. The two methods differ in how the guards are expressed (code against prose), in how a pattern is recognised (a fixed expression against the model’s reading of the quotes), and in determinism: the rules always return the same answer for the same record.

#### E. Evaluation

F1 is the standard metric used to report results of seizure frequency extraction, but authors differ in the granularity at which a correct frequency label is defined. To allow comparison with Gan *et al.* [3], this paper adopts their two evaluation categories: a fine-grained *purist* category and a coarser *pragmatic* category. If the gold label is *twice a week* and the prediction is *once a week*, that is a correct *pragmatic* answer, since both fall in the *frequent* category, but an incorrect *purist* answer. Table VI lists the categories in each.

Micro-F1 is reported for both because the categories are badly imbalanced (some have over 100 letters, some fewer than 10; Table VI) and micro-F1 gives each letter equal weight. Each letter has exactly one gold label and one prediction, so

TABLE IV: Comparison with previously reported results

| Author         | Year | Method         | Purist F1 | Pragmatic |
|----------------|------|----------------|-----------|-----------|
| Holgate et al. | 2024 | Prompted LLM   | -         | 0.73      |
| Holgate et al. | 2025 | Fine-tuned LLM | 0.68      | 0.81      |
| Gan et al.     | 2026 | Fine-tuned LLM | 0.79      | 0.85      |
| This paper     | 2026 | Hybrid         | 0.80      | 0.84      |

Gan et al. and this paper report scores on the same 300 real letters. Holgate et al. used different data.

TABLE V: Purist micro-F1 on 300 real patient letters

| Method                              | Model             | Purist F1 |
|-------------------------------------|-------------------|-----------|
| Fine-tuned on synthetic letters [3] | Qwen2.5-14B       | 0.79      |
| Hybrid (this paper)                 | DeepSeek V4 Flash | 0.80      |
| LLM-only (this paper)               | DeepSeek V4 Flash | 0.77      |

micro-F1 here equals the share of letters answered correctly. The more precise *purist* F1 is the primary metric.

All prompts, schema and rules were developed on the 750-letter development set, using qualitative error analysis to find failure modes and test changes. The pipeline is primarily evaluated on the held-out test set of 450 letters. Results from the fine-tuned model in Gan *et al.* [3] were compared with this pipeline using the same set of 300 real patient clinic letters.

The primary model is Gemini 3.7 Flash. It is the only model used for the prompt ablations and for the temperature and thinking-level ablations. Six models were evaluated (Section IV-B). DeepSeek was the only model used on the set of real patient letters. Elsewhere, Hybrid results are from Gemini at temperature 0 with low thinking effort.

## IV. RESULTS

### A. Benchmark Performance

Hybrid achieved a *purist* F1 of 0.80 and LLM-only achieved 0.77 on the 300 real patient clinic letters, demonstrating the two-stage pipeline can achieve comparable results to the fine-tuned LLM [3]. These results were achieved using a DeepSeek model that was small enough that it could fit on the university’s High Performance Computing (HPC) system and run locally; this is important as it shows that models small enough to fit on local hospital servers, and therefore able to process patient data without sending it off-site, can achieve strong performance with this method.

The pipeline is particularly effective at extracting high-frequency events. On the synthetic test set Gan *et al.* report unknown as their highest scoring category (0.89), with daily (0.70) and more than weekly (0.68) much lower. In this pipeline, daily (0.92) and more than weekly (0.91) are among the best performing categories, well above unknown (0.74). The full breakdown across *purist* and *pragmatic* categories is in Table VI.

### B. Model Comparison

Both methods were implemented with six models (Table VII; Figure 3 shows the Hybrid lift). Hybrid beat LLM-only on every model. Rules add the most where the

TABLE VI: Hybrid Purist and Pragmatic F1 on 450 synthetic letters

| Purist                                      | <i>n</i> | Purist F1 | Pragmatic F1       |
|---------------------------------------------|----------|-----------|--------------------|
| Daily                                       | 41       | 0.92      | Frequent: 0.95     |
| More than weekly, less than daily           | 123      | 0.91      |                    |
| Once a week                                 | 5        | 0.89      |                    |
| More than monthly, less than weekly         | 58       | 0.90      |                    |
| Once a month                                | 18       | 0.94      | Infrequent: 0.82   |
| More than every 6 months, less than monthly | 55       | 0.81      |                    |
| Once every 6 months                         | 1        | 0.00      |                    |
| Less than once every 6 months               | 6        | 0.60      |                    |
| Unknown                                     | 76       | 0.74      | Unknown: 0.74      |
| Seizure free                                | 67       | 0.90      | Seizure free: 0.90 |

Rows with  $n < 10$  shown in grey.

extraction call is weakest: Luna gains +0.10 from the provisional answer, DeepSeek +0.08, while Gemini and Grok gain +0.07 and +0.06. The second LLM call helps the stronger models (+0.06 on Gemini, +0.05 on Grok and Luna, +0.03 on DeepSeek) and *lowers* the local models (Qwen loses 0.05 and Gemma 0.04 versus their own provisional answer). Those two models already quote less exactly and need more format repair (Table VIII); asking them to re-apply the policy from the record introduces errors that the recorded rules do not.

Purist F1 closely tracks general medical capability as measured by Artificial Analysis’ Healthcare and Medical Index. The models cluster into three groups: strong (Gemini 0.86, Grok 0.85), medium (DeepSeek 0.82, Luna 0.79) and weak (Qwen 0.76, Gemma 0.72), which largely reflect model size and cost.

Table VIII presents a wider view of performance: how often did the models provide exact evidence and adhere to the exact schema requirements. The four hosted models quoted an exact substring of the letter as evidence for the selected event in 99% or more of letters; the two local models paraphrased more often (Qwen 97%, Gemma 93%) and needed more JSON format repair. An exact span is not proof of a correct answer, but it is proof of a verifiable answer. Overall the models demonstrated they are capable of populating rich schemas with strict requirements consistently and reliably.

Table IV also lists the two earlier papers that used the same pragmatic metric on different data. Holgate *et al.* [10] ran Llama 2 on local hospital servers with eight Nvidia V100 GPUs and reached a pragmatic F1 of 0.73 without fine-tuning. In Hybrid, Qwen reached a pragmatic F1 of 0.80 running locally on a laptop with a single GPU. This shows that the design can execute on local hardware under the same synthetic task conditions.

### C. Extraction Prompt Ablations

Table IX removes one component of the extraction prompt at a time and evaluates the Hybrid pipeline at both stages, so the effect of each component can be seen on the provisional

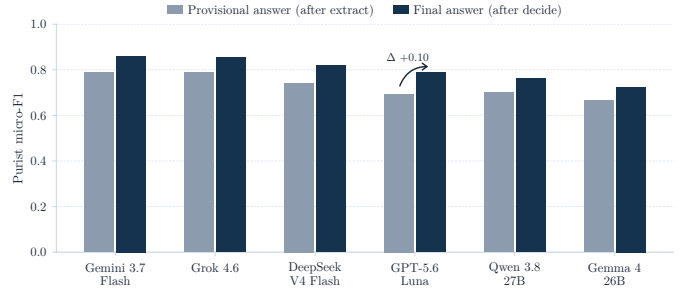


Fig. 3: Hybrid Purist F1 for six models: the LLM’s provisional answer after `extract` and the final answer after rule-based `decide`. Every model gains at `decide`; Luna gains most (0.69 to 0.79). Table VII adds the second-call scores.

TABLE VII: Purist F1 after extract and after each method

| Model             | Provisional | Hybrid              | LLM-only     |
|-------------------|-------------|---------------------|--------------|
| Gemini 3.7 Flash  | 0.79        | <b>0.86</b> (+0.07) | 0.85 (+0.06) |
| Grok 4.6          | 0.79        | 0.85 (+0.06)        | 0.84 (+0.05) |
| DeepSeek V4 Flash | 0.74        | 0.82 (+0.08)        | 0.77 (+0.03) |
| GPT-5.6 Luna      | 0.69        | 0.79 (+0.10)        | 0.74 (+0.05) |
| Qwen 3.8 27B      | 0.70        | 0.76 (+0.06)        | 0.65 (−0.05) |
| Gemma 4 26B       | 0.66        | 0.72 (+0.06)        | 0.62 (−0.04) |

The provisional answer reflects the model’s answer after the `extract` phase which both Hybrid and LLM-only methods use as input. Deltas are versus the provisional answer.

TABLE VIII: Six-model comparison: performance, exact evidence and schema adherence

| Model             | Purist      | Prag.       | Exact ev. | Repairs |
|-------------------|-------------|-------------|-----------|---------|
| Gemini 3.7 Flash  | <b>0.86</b> | 0.88        | 99.8%     | 0       |
| Grok 4.6          | 0.85        | <b>0.89</b> | 100%      | 0       |
| DeepSeek V4 Flash | 0.82        | 0.85        | 99.6%     | 2       |
| GPT-5.6 Luna      | 0.79        | 0.82        | 98.9%     | 0       |
| Qwen 3.8 27B      | 0.76        | 0.81        | 96.7%     | 3       |
| Gemma 4 26B       | 0.72        | 0.77        | 92.9%     | 15      |

Exact ev.: the quoted span for the selected event is an exact substring of the letter. Repairs: letters needing JSON dialect repair.

answer and after rules have applied the policy. Removing the worked examples, the closed list of allowed label forms, or the evidence obligation each lowers the provisional answer by -0.02 and the final answer by -0.04. Their individual effect is roughly equal. What this shows is while each component has some effect, the prompt design is fairly robust to any one change.

The result for the evidence obligation is of particular interest. Examples and label forms are designed to help the LLM choose the right answer by describing what the right answer looks like. The evidence obligation is different; it is an additional layer of task complexity to include the exact evidence quotation along with the answer label. Including the evidence helps at the `decide` stage because the rules can use the exact quotes to determine the correct answer even if the LLM provides the wrong label. However, it also appears

TABLE IX: Extraction prompt ablations on 450 synthetic letters (Hybrid)

| Extraction prompt                  | Provisional answer | Final answer |
|------------------------------------|--------------------|--------------|
| Full prompt                        | 0.79               | <b>0.86</b>  |
| Without examples                   | 0.77               | 0.82         |
| Without evidence obligation        | 0.77               | 0.82         |
| Without closed allowed-label forms | 0.77               | 0.81         |

Each row changes only the extraction prompt; the same rules apply the policy at decide. Effects are not additive.

to help in the initial `extract` phase (+0.02) by encouraging the LLM to ground its answer in the evidence. This shows that there is no trade-off between schema complexity and accuracy; a more complex schema with more stringent requirements produced more evidence and more accurate answers.

#### D. Model Configuration

Raising the reasoning effort level did not improve performance. Gemini’s `purist`-F1 score was 0.86 at low and 0.84 at high; a paired difference of +0.02, 95% CI -0.01 to 0.04;  $p = 0.31$ . Extra reasoning adds cost and latency without improving extraction. The temperature setting also had no significant effect: 0.86 at temperature 0 versus 0.84 at temperature 1 (paired difference +0.02, 95% CI -0.01 to 0.04;  $p = 0.20$ ). The task is tightly defined by an explicit schema, which limits the expected variability by design and partly explains why neither setting mattered.

#### E. Error Analysis

The worst performing category was `unknown` with an F1 score of 0.74. 69% of the errors (37/54) are incorrect answers of `unknown`. The full error distribution is shown in the confusion matrix (Figure 4).

The three most common errors are `infrequent`  $\rightarrow$  `unknown` (16), `frequent`  $\rightarrow$  `unknown` (13), and `seizure free`  $\rightarrow$  `unknown` (8). The errors reflect a cautious approach when the evidence is ambiguous, for example the model recognised the phrase ‘these episodes crop up most shifts’ as the primary seizure frequency description but refused to make an assumption about how many shifts the person works. The gold-standard label was `multiple per week` but the model chose `unknown`. This cautious assessment is justifiable, and because the exact evidence is preserved, downstream users can see the proposed answer and draw their own conclusion from the quoted span.

#### F. Experimental Environment

Table X summarises the hardware, software and model settings used. DSPy co-ordinated the model calls and required structured JSON output, Pydantic validated each record against the schema, and LiteLLM delivered API calls to Ollama and the model routers. Full specifications, API settings and replay artefacts are in the supporting materials.

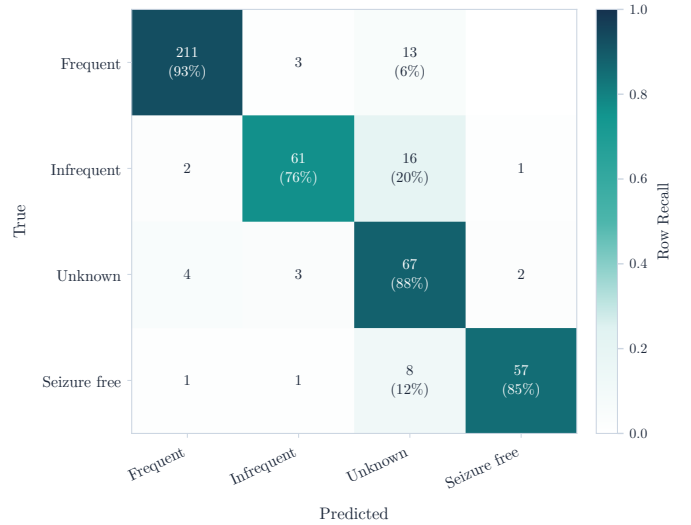


Fig. 4: Hybrid confusion matrix on the pragmatic categories (test set). Most errors are incorrect predictions of `unknown`.

TABLE X: Experimental environment

| Component      | Setting                                                                                                                                                                      |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local models   | Qwen 3.8 27B and Gemma 4 26B served by Ollama 0.32.15 on a Dell XPS 16 9640 (Windows 11, Intel Core Ultra 9 185H, 32 GB RAM, NVIDIA RTX 4070 Laptop GPU, 8 GB VRAM)          |
| Hosted models  | Gemini 3.7 Flash, Grok 4.6, GPT-5.6 Luna and DeepSeek V4 Flash, called from a Mac mini M1 (16 GB, macOS 26.5.2) directly (OpenAI, DeepSeek) or via OpenRouter and AI Gateway |
| Model settings | Temperature 0 (Luna accepts only 1); thinking or effort level low.                                                                                                           |
| Software       | Python 3.14.4, DSPy 3.2.1, Pydantic 2.13.4, LiteLLM 1.87.1                                                                                                                   |

## V. DISCUSSION

This paper shows that a two-stage pipeline can extract an epilepsy patient’s current seizure frequency pattern with comparable accuracy to a fine-tuned model [3]. A single prompt designed to extract every candidate with exact evidence, followed by rules or a second call that decide from that evidence, matched that benchmark. On every model the rules beat a second LLM call, and this pattern translated from the synthetic dataset to the real patient data.

The method is particularly effective at identifying the categories that mark the greatest disease burden. F1 scores of 0.92 for daily and 0.91 for more than weekly are around 0.2 higher than the category scores previously reported [3], although on a different sample. Structured seizure frequency of this kind is intended for later research uses: cohort identification for trials, retrospective and longitudinal analyses of treatment response, and outcome variables for predictive models [13], which together can widen the scope of real-world evidence [17].

Past work has typically tasked the LLM with producing a

single answer in a fully normalised form [3]. That discards the semantic depth of the narrative, which clinicians value because it conveys reasoning, uncertainty and vague impressions, and those nuanced observations are more accurate and reliable [18]. Current LLMs can populate a richer record that preserves them: every candidate with its quoted span, the policy applied, and the label it produced. The ablations show that this record is also what the accuracy depends on. A more explainable system is often a more useful system in a clinical setting [14], and here it is also a more accurate system.

There are several limitations. A single model (DeepSeek) was run on an external test set of patient data and showed strong transfer, but all other evaluations in the research used synthetic data. Synthetic data has been shown to produce more positive results in prior papers as it contains fewer typos, idiosyncrasies and a more consistent structure than real data. Past results have also shown that performance on one corpus of real patient data is not evidence that it will translate to other hospital templates or clinician writing styles [8]. Results show that the model can produce accurate answers with exact evidence quoted from the letter, but it does not evaluate whether the evidence quoted entailed the result. Future research should examine this in more detail with expert evaluations of the evidence collected. Finally, the method was applied to one narrow task. The same method has been configured for a multi-fact epilepsy inventory task [7], to be reported separately, and future work should evaluate whether the design generalises to other clinical IE tasks and whether a more general framework, in which an LLM chooses how to represent facts and which policy to apply, adds value as reported in oncology [19] and cardiology [20].

## VI. AI ETHICAL CONSIDERATIONS

The majority of evaluations were run using synthetic clinic letters released by Gan *et al.* [3] that were purposefully built as a privacy-preserving tool. Those letters are modelled on real UK epilepsy letters but do not contain identifiable patient records. The author of Gan *et al.* [3] used this pipeline on their set of 300 real patient letters that they acquired prior approval for. The pipeline was executed on the university’s HPC system using the DeepSeek model which demonstrated this method can achieve strong performance using systems that can run on local servers. This is important as hospitals prefer processing patient data on their own servers as it is highly sensitive [6].

LLMs can invent evidence, so careful consideration was given to the prompt instructions and schema design. The model is required to quote exact evidence spans from the letter, and every later change to the answer is made by a named rule or a second call applied to that quoted evidence. A reviewer can therefore trace any final label back to the text that supports it and to the policy that decided it [14]. This transparency matters more than headline accuracy in a clinical setting, where an unexplained answer cannot be safely acted on.

The letters contain no demographic attributes, so unequal performance across patient groups cannot be tested, and the synthetic corpus may not reflect the language of every clinic.

Bias can enter through the models, the prompts and the decision rules; for example, the cautious default to `unknown` (Section IV-E) would systematically under-count patients whose seizures are described vaguely. Rules make the decision policy inspectable and changeable by local clinicians, but they do not remove the need for local clinical review and fairness testing on real records before any operational use.

## VII. CONCLUSION

Seizure frequency is a core outcome measure in epilepsy, recorded almost exclusively in narrative clinic letters. This paper proposes a pipeline in which one LLM call extracts every candidate fact with its quoted evidence in a canonical form, then a decision applies a fixed policy to those candidates.

The Hybrid (Purist F1: 0.80) and LLM-only (0.77) delivered comparable performance to previously reported results for a fine-tuned LLM (0.79) on a set of 300 clinic letters for real patients. On the synthetic dataset the provisional answer reaches 0.79; applying the decision policy to the evidence it found adds a further +0.06 to +0.07 on Gemini. Across six models, rules at `decide` always beat a second LLM call. Four of the six models extracted exact evidence from the letter with 99%+ accuracy, including a model small enough to fit on some hospital servers.

Automatically extracting seizure frequency patterns from EHRs could make this outcome measure available for large-scale clinical research: cohort identification, retrospective studies and predictive modelling. Clinical review, workflow evaluation and a broader set of clinical information extraction tasks remain before any of those uses can be validated.

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